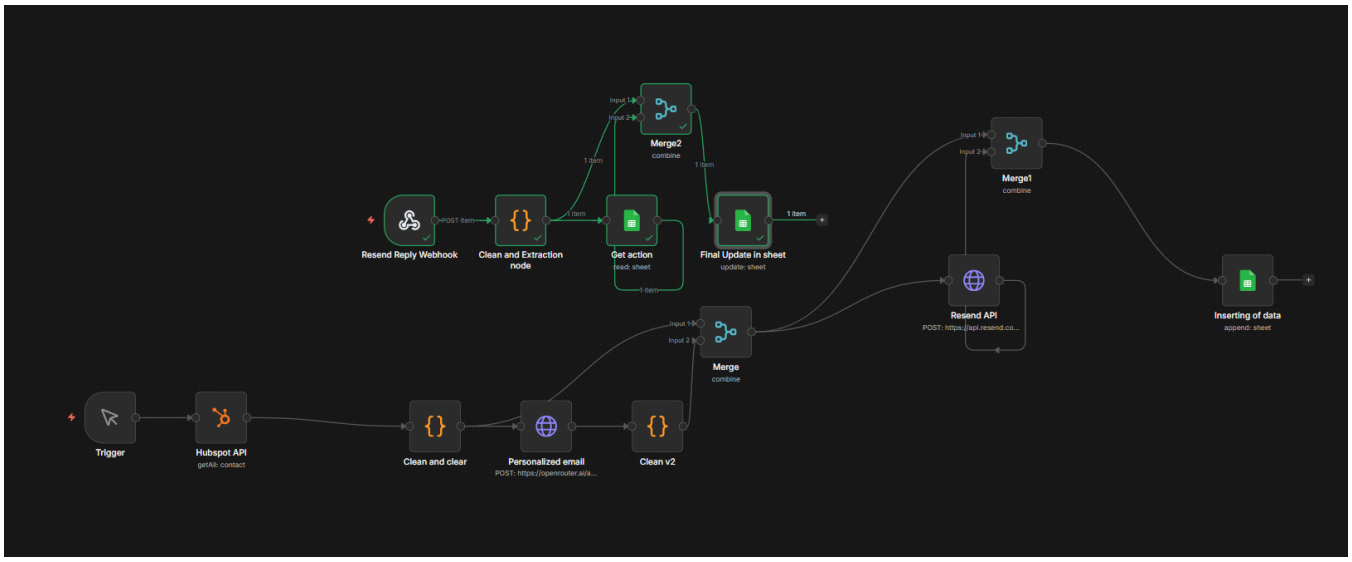
GTM Assignment 2

Anakin AI Outreach Pipeline — Workflow Documentation

13	2	4	3
Total Nodes	Pipeline Flows	Integrations	Merge Points

Internal Use · GTM Automation · March 2026



01 Overview

This n8n workflow automates Anakin AI's cold outreach end-to-end — from HubSpot contact extraction through personalised email generation, delivery, and reply tracking — all without manual intervention.

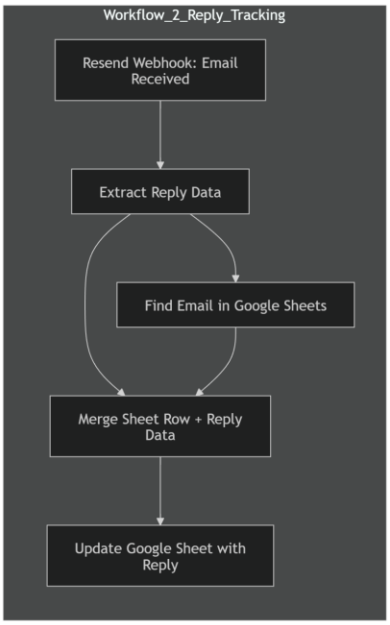
Two Parallel Flows

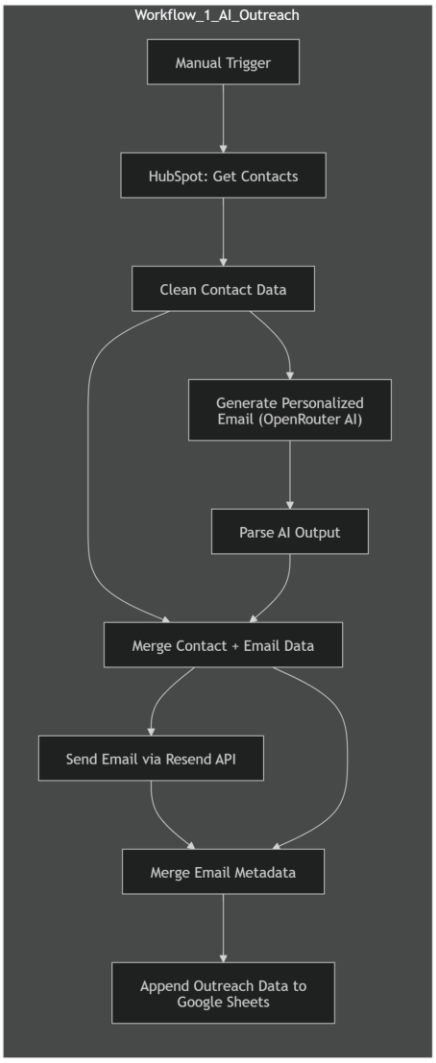
Flow	Name	Trigger
A	Outreach Pipeline	Manual (click Execute Workflow)
B	Reply Tracking Pipeline	POST Webhook from Resend

Stack Summary

Tool	Role	Auth
n8n (self-hosted)	Orchestration	—
HubSpot	Contact source (CRM)	App Token
OpenRouter → GPT-4o-mini	Email personalisation	Bearer Token (HTTP Header)
Resend	Email delivery + reply webhook	HTTP Header Auth
Google Sheets	CRM output + reply log	OAuth2

Workflow Diagram:-





02 Data Flow

Flow A — Outreach

```
Trigger
  └─▶ Hubspot API  (fetch all contacts)
        └─▶ Clean and clear  (flatten HubSpot structure)
              ├─▶ Personalized email  (OpenRouter / GPT-4o-mini)
              │      └─▶ Clean v2  (parse subject + body)
              │            └─▶ Merge ◀──────────────────────────┐
              └──────────────────────────────────────────────────┘
                                  │
                        ┌─────────┴─────────┐
                        ▼                   ▼
                  Resend API            Merge1 ◀── Resend API response
                  (send email)              │
                                            ▼
                                  Inserting of data
                                  (append row to Sheets)
```

Flow B — Reply Tracking

```
Resend Reply Webhook  (POST /resend-replies)
  └─▶ Clean and Extraction node  (parse reply payload)
        ├─▶ Get action  (lookup row in Sheets by email)
        │      └─▶ Merge2 ◀──────────────────────────────┐
        └──────────────────────────────────────────────────┘
                          │
                          ▼
                Final Update in sheet
                (update status + reply_time by row_number)
```

03 Flow A — Outreach Pipeline (Nodes 1–9)

#	Node Name	n8n Type
1	Trigger	Manual Trigger

Purpose	Starts Flow A on demand. Click Execute Workflow in n8n to begin a full outreach run.
Upgrade tip	Swap for a Schedule Trigger to run automatically (e.g. nightly at 08:00).

#	Node Name	n8n Type
2	Hubspot API	HubSpot — Get All Contacts

Credential	HubSpot App Token
Fetch mode	returnAll: true — fetches every contact in one operation
Properties	hs_linkedin_url, firstname, lastname, jobtitle, company, about, linkedin_headline
Output	One item per contact, streamed individually through the rest of the pipeline

#	Node Name	n8n Type
3	Clean and clear	Code (JavaScript)

Purpose	Flattens the nested HubSpot API response into a simple object
Key mapping	bio ← linkedin_headline · email ← identity-profiles[0].identities (type=EMAIL)
Null safety	All fields default to empty string if missing — pipeline never errors on sparse contacts
Output fields	firstname, lastname, email, linkedin_url, jobtitle, company, about, bio
Branches	Two outputs: Branch 1 → Personalized email | Branch 2 → Merge input 0

#	Node Name	n8n Type
4	Personalized email	HTTP Request (POST)

URL	https://openrouter.ai/api/v1/chat/completions
Model	openai/gpt-4o-mini (via OpenRouter)
Credential	openRouterApi Bearer Token
System prompt	You are a GTM automation specialist writing personalized cold outreach emails.
Prompt inputs	firstname, lastname, jobtitle, company, bio (LinkedIn headline), about
Instructions	Reference recipient role/expertise · connect to Anakin AI · under 90 words · friendly CTA · include Subject: line
Raw output	choices[0].message.content — passed directly to Clean v2 for parsing

#	Node Name	n8n Type
5	Clean v2	Code (JavaScript)

Purpose	Parses the raw LLM response into two clean fields: subject and email_body
Subject logic	Regex match on /Subject:\s*(.*)/i · fallback: 'AI automation idea'
Body logic	Strips the Subject: line and boilerplate sign-offs (e.g. 'Best, [Your Name]')
Output fields	firstname, email, company, subject, email_body
Feeds into	Merge input 1

#	Node Name	n8n Type
6	Merge	Merge — Combine by Position

Input 0	Clean and clear — full contact record
Input 1	Clean v2 — parsed subject + email_body
Mode	combineByPosition — requires same item count on both branches
Output fields	All contact fields + subject + email_body in one object
Branches	Branch 1 → Resend API (send) | Branch 2 → Merge1 input 0 (log)

#	Node Name	n8n Type
7	Resend API	HTTP Request (POST)

URL	https://api.resend.com/emails
Credential	HTTP Header Auth (Resend API key as Authorization: Bearer)
Batch size	1 — one API call per contact to stay within Resend rate limits
From	Aryan <hello@aryancore.in> — update this field to change sender identity
Body format	HTML: greeting + email_body with \n → conversion + hardcoded signature block
Returns	Resend email ID (id) — captured in Merge1 for logging

#	Node Name	n8n Type
8	Merge1	Merge — Combine by Position

Input 0	Merge node — full contact + email record
Input 1	Resend API response — contains the Resend-assigned email id
Purpose	Makes the Resend email ID available alongside contact data for the CRM write
Feeds into	Inserting of data

#	Node Name	n8n Type
9	Inserting of data	Google Sheets — Append Row

Sheet	ANAKIN OUTREACH (gid=0)
Spreadsheet	GTM Assignment 2 (ID: 1ivETqLu7XA2PpUbEtbL0FMNisw05-knO-RjpsPWVLwU)
Credential	Google Sheets OAuth2
Columns written	id (Resend), firstname, lastname, email, linkedin_url, jobtitle, company, about, bio, subject, email_body
Match key	id — the Resend email ID. Used by Flow B to match replies to rows

04 Flow B — Reply Tracking Pipeline (Nodes 10–14)

#	Node Name	n8n Type
10	Resend Reply Webhook	Webhook — POST

Path	/resend-replies
Webhook ID	8d0e1070-9db7-4ecc-bd14-53f31e2025a9
Registration	Add to Resend dashboard under Domains → Webhooks as reply event URL
Full URL	https://<your-n8n-host>/webhook/resend-replies
Fires when	A recipient replies to any email sent by Resend from this account

#	Node Name	n8n Type
11	Clean and Extraction node	Code (JavaScript)

Purpose	Extracts reply metadata from the nested Resend event payload
Source path	Fields come from body.data — Resend wraps event data one level deep
Output fields	email (data.from), subject (data.subject), resend_email_id (data.email_id), message_id, received_at (body.created_at), event_type (body.type)
Branches	Branch 1 → Get action (row lookup) | Branch 2 → Merge2 input 0

#	Node Name	n8n Type
12	Get action	Google Sheets — Get Row(s) by Filter

Sheet	ANAKIN OUTREACH
Filter	email = {{ $json.email }} — matches the replier's email address
Returns	The original outreach row including row_number — critical for the update step
Feeds into	Merge2 input 1

#	Node Name	n8n Type
13	Merge2	Merge — Combine by Position

Input 0	Clean and Extraction node — parsed reply metadata
Input 1	Get action — original row data including row_number
Purpose	Combines reply event data with the row reference needed to perform the update
Feeds into	Final Update in sheet

#	Node Name	n8n Type
14	Final Update in sheet	Google Sheets — Update Row

Sheet	ANAKIN OUTREACH
Match key	row_number — read-only field auto-populated by Sheets; uniquely identifies each row
Fields updated	id ← resend_email_id · reply_time ← received_at · status ← $json.status
Note	reply_subject is currently mapped as empty — add $json.subject here if subject tracking is needed

05 Google Sheets Schema

All data is written to the ANAKIN OUTREACH tab of spreadsheet ID 1ivETqLu7XA2PpUbEtbL0FMNisw05-knO-RjpsPWVLwU. Replace this ID in all three Sheets nodes if using a different spreadsheet.

Column	Written By	Description
firstname	Flow A — Node 9	First name from HubSpot
lastname	Flow A — Node 9	Last name from HubSpot
email	Flow A — Node 9	Primary email from HubSpot identity profile
linkedin_url	Flow A — Node 9	hs_linkedin_url from HubSpot
jobtitle	Flow A — Node 9	Job title from HubSpot
company	Flow A — Node 9	Company name
about	Flow A — Node 9	About field from HubSpot
bio	Flow A — Node 9	LinkedIn headline (linkedin_headline)
subject	Flow A — Node 9	AI-generated subject line (Clean v2)
email_body	Flow A — Node 9	AI-generated email body (Clean v2)
id	Flow A — Node 9 / Flow B — Node 14	Resend email ID; updated to resend_email_id on reply
status	Flow B — Node 14	Reply status from Resend webhook event
reply_subject	Flow B — Node 14	Subject of reply (currently empty — add $json.subject to populate)
reply_time	Flow B — Node 14	Timestamp of reply (body.created_at from Resend)
row_number	Auto (Sheets)	Read-only row index used as update key by Flow B

06 Error Handling & Known Limitations

Issue	Impact	Recommended Fix
All HubSpot contacts fetched every run	Duplicate sends if run twice	Add IF node to skip contacts where email_body is already populated in Sheets
No batch size cap on HubSpot pull	API rate limit risk at scale	Add a Limit node after Hubspot API node
Merge uses combineByPosition	Misalignment if one branch errors mid-run	Monitor execution logs; add error routing on Personalized email node
No retry on Personalized email	Contact silently dropped on OpenRouter timeout	Set onError: continueRegularOutput; add 3-attempt retry policy
Missing email on HubSpot contact	Resend API call fails for that item	Add IF node before Resend API to filter empty emails
No suppression / unsubscribe list	CAN-SPAM and GDPR compliance risk	Add suppression check before Resend API; include unsubscribe link in HTML template
reply_subject mapped as empty	Reply subject not captured in Sheets	Map $json.subject into reply_subject field in Final Update node

07 Production Checklist

- Verify sending domain aryancore.in is warmed with DKIM and SPF configured in Resend
- Add deduplication check to skip contacts already present in ANAKIN OUTREACH
- Register Resend webhook URL in Resend dashboard: https://<your-n8n-host>/webhook/resend-replies
- Test with a single pinned contact before executing a full HubSpot pull
- Add onError: continueRegularOutput on the Personalized email and Resend API nodes
- Cap HubSpot pull with a Limit node during initial testing
- Add an unsubscribe link to the HTML email template in the Resend API node
- Implement suppression list check before every send cycle
- Map $json.subject into reply_subject in Final Update in sheet

08 Node Quick Reference

#	Node Name	Type	Function
1	Trigger	Manual Trigger	Starts Flow A
2	Hubspot API	HubSpot	Fetches all CRM contacts
3	Clean and clear	Code (JS)	Flattens HubSpot response
4	Personalized email	HTTP Request	LLM email generation via OpenRouter
5	Clean v2	Code (JS)	Parses subject + body from LLM output

#	Node Name	Type	Function
6	Merge	Merge	Joins contact record + email content
7	Resend API	HTTP Request	Sends email via Resend
8	Merge1	Merge	Joins merged record + Resend response
9	Inserting of data	Google Sheets	Appends row to ANAKIN OUTREACH tab
10	Resend Reply Webhook	Webhook	Receives reply events (Flow B entry)
11	Clean and Extraction node	Code (JS)	Parses Resend reply payload
12	Get action	Google Sheets	Looks up original row by email
13	Merge2	Merge	Joins reply data + row lookup
14	Final Update in sheet	Google Sheets	Updates status + reply_time by row_number